

Andrew MM Matheson

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Summary

I use a combination of behavioural, transcriptomic and physiological techniques in newts to understand how the nervous system adapts to changing environments within a species and across evolution.

Education

PhD	New York University , Neuroscience and Physiology	New York, NY, USA
	- Thesis - The Neural Circuits that Support Wind-Guided Olfactory Navigation, advised by Katherine I Nagel	Jan 2015 – Jan 2021
BSc	McGill University , Computer Science and Biology	Montreal, QC, Canada
	Researched vocal communication using songbirds in Jon T Sakata's lab	Jan 2011 – Jan 2015

Experience

Tosches Lab, Department of Biological Sciences, Columbia University , Postdoctoral Research Scientist	New York, NY
Investigating neural mechanisms of environmental plasticity and sensory processing in the Iberian ribbed newt.	Jan 2021 – present 5 years 6 months

Publications

Iberian ribbed newts	Jan 2025
Matheson AMM, Chua NJ, Tosches MA 10.1016/j.cub.2024.11.062 (Current Biology)	
A neural circuit for wind-guided olfactory navigation	Jan 2022
Matheson AMM, Lanz AJ, Currier TA, Licata AM, Syed MH, Nagel KI 10.1038/s41467-022-32247-7 (Nature Communications)	
A population of central complex neurons promoting airflow-guided orientation in <i>Drosophila</i>	Jan 2020
Currier TA, Matheson AMM, Nagel KI 10.7554/eLife.61510 (eLife)	
Central neurons encoding wind direction in <i>Drosophila</i>	Jan 2019
Suver MP, Matheson AMM, Sarkar S, Damiata M, Schoppik D, Nagel KI 10.1016/j.neuron.2019.03.012 (Neuron)	
Cortical control of vocal interaction in neotropical singing mice	Jan 2019
Okobi DE, <i>Banerjee A</i> , Matheson AMM, Phelps SM, Long MA 10.1126/science.aau9480 (Science)	
Relationship between the sequencing and timing of vocal motor elements in birdsong	Jan 2015
Matheson AMM, Sakata JT 10.1371/journal.pone.0143203 (Plos One)	

Awards

Doupe Fellowship	Jan 2026
Awarded by the McKnight Endowment Fund for Neuroscience.	

Young Investigator Award

Jan 2026

Awarded by the International Society for Neuroethology. Award recognizing early-career achievements in neuroethology.